

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

__show ip route ospf__

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? __S0/0/0__

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? __129__

Does R2 show up as an OSPF neighbor on R1? __Yes__

Does R2 show up as an OSPF neighbor on R3? __No__

What does this information tell you?

__Setting the interface connected to the LAN (G0/0) on R2 as passive prevents OSPF hello packets from being sent/received on that interface. This stops OSPF neighbor adjacencies from forming on the LAN segment (no unnecessary hellos or adjacencies with potential unauthorized devices or switches/hosts). However, the connected network (192.168.2.0/24) is still advertised into OSPF via other adjacencies (e.g., serial links to R1 and R3), so routing to it remains functional through the alternative path (via R2's serial links). R3 loses direct adjacency to R2 on the "passive" side but can still reach the network via the triangle topology.__

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

__R2(config)# router ospf 1, R2(config-router)# no passive-interface s0/0/1__ Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? __S0/0/1__

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

__65 (cost 64 on R3-R2 serial link + cost 1 on R2's G0/0 LAN interface)__

Is R2 listed as an OSPF neighbor to R3? __Yes__

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default reference bandwidth is 100 Mbps, so any interface ≥ 100 Mbps gets a cost of 1. This causes poor differentiation between link speeds. Changing it allows more accurate cost calculation and better path selection for faster modern links.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to 192.168.3.0/24: R1 S0/0/1 + R3 G0/0 ($781+1=782$). The cost to 192.168.23.0/30: R1 S0/0/1 and R3 S0/0/1 ($781+64=845$).

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1,562. Each serial link now has a cost of 781, and the route to the 192.168.23.0/24 network travels over two serial links. $781 + 781 = 1,562$

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF will choose the route with the least accumulated cost. The route with the lowest accumulated cost is: R1-S0/0/0 + R2-S0/0/1 + R3-G0/0, or $781 + 781 + 1 = 1,563$. This metric is smaller than the accumulated cost of R1-S0/0/1 + R3-G0/0, or $1565 + 1 = 1,566$.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

__Router ID must be unique in the OSPF domain (used in LSAs, DR/BDR election, troubleshooting). If not manually set (loopback preferred), it uses highest IP (can change on reload if interfaces flap), causing instability, duplicate RID conflicts, or election issues. _____

2. Why is the DR/BDR election process not a concern in this lab?

__All links are serial point-to-point (default OSPF network type point-to-point). No multi-access segments, so no DR/BDR election occurs — routers form direct full adjacencies without DR/BDR optimization. _____

3. Why would you want to set an OSPF interface to passive?

__To prevent sending/receiving OSPF hello packets on that interface, reducing CPU overhead, unnecessary traffic, and security risks. The connected network is still advertised into OSPF via other interfaces. _____

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

__The use of areas (multi-area OSPF). Area 0 (backbone) connects other areas; this hierarchy limits LSA flooding, reduces SPF calculations per router, and allows summarization — enabling large, scalable networks. _____

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

__network 10.0.0.0 0.255.255.255 area 0 _____

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

~~__The RIP route (metric 4).~~ Administrative distance (AD) decides first: RIP AD=120, OSPF AD=110, EIGRP AD=90 (internal). Lowest AD wins (EIGRP 90), but wait — actually EIGRP has lowest AD=90, so EIGRP route (4039043) is preferred over OSPF (110) and RIP (120). Correction: EIGRP has AD 90, so the EIGRP route is installed. _____